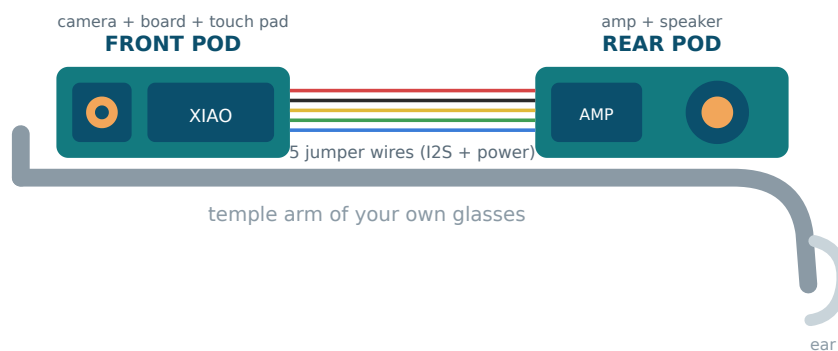


Vocalens

Assembly Booklet

Two clip-on pods, six wires, zero purchases. Build it in the order printed here and every step is testable before you commit to the next.



The finished device, worn. Front pod near the hinge, rear pod behind the ear.

WHAT THIS BOOKLET ASSUMES YOU ALREADY HAVE

PRINTED (STEP 0)

- **x1** frontboard_base — TPU
- **x1** frontboard_lid — PETG
- **x1** rearaudio_base — TPU
- **x1** rearaudio_lid — PETG
- **x1** fit_test — PLA, already tuned

ELECTRONICS

- **x1** XIAO ESP32S3 **SENSE**
- **x1** OV2640 camera + ribbon
- **x1** Audio Converter & Amplifier
- **x1** Speaker (round, ~28mm)
- **x1** Pair of glasses to clip onto

Do not start until fit_test grips your glasses properly. Everything downstream assumes the snap-fit is tuned. If the clip is loose or splays the arm, go back and adjust `SNAP_GAP` in `vocalens_pod.scad` and reprint — it's a three-minute print and it saves you an evening.

THE 7-PIN HEADERS: DON'T FIT THEM

The XIAO ESP32S3 Sense is **two boards**: the XIAO itself, and the Sense expansion board underneath carrying the camera connector, PDM microphone and microSD slot, mated through a board-to-board connector. The supplied 2×7 headers insert from the top and their pins protrude **downward** — into exactly the space the expansion board occupies. That's why they won't go in. It isn't user error and nothing is broken.



Left: why the headers jam. Right: what to do instead.

Why skip the headers at all? You need six connections, not fourteen — and the pods are only **10 mm tall**. That number is the entire reason they clear the wearer's head. A header strip plus its socket spends a chunk of that budget on pins you'll never use. Solder to the castellated pads instead.

IF YOU DO NEED TO SEPARATE THE TWO BOARDS

- 1 Unplug the camera ribbon first — release the dark flip-latch on the connector, then slide the ribbon out. Never pull it while latched.
- 2 Pry **evenly from the two short ends**, a little at a time, alternating.
- 3 Do not twist, and never lever against the camera connector — the most fragile thing on the board.

Before you power anything on: set `PSRAM` to **OPI PSRAM** in your build settings. If you skip this the camera silently fails to initialise and you will lose an hour to a bug that isn't in your code.

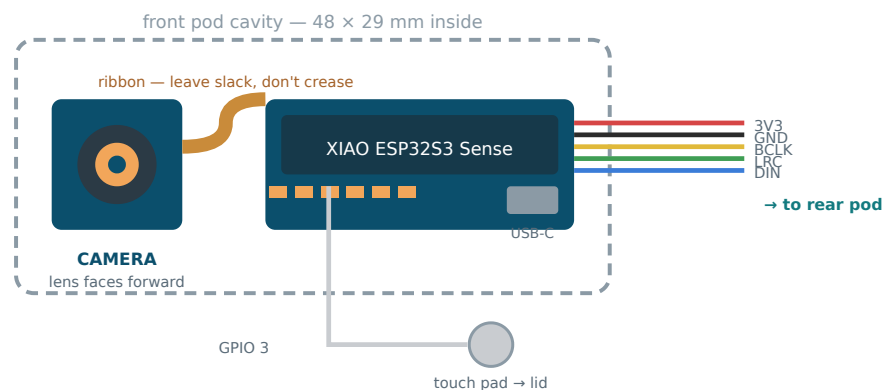
TIN THE SIX PADS NOW

Blob a little solder on each of the six pads before attaching any wires — it turns every later joint into a one-second touch rather than a three-handed wrestle inside a 10 mm cavity.

PAD	SIGNAL	WIRE COLOUR
3V3	Power out to amp	Red
GND	Ground	Black
D7 / GPIO 7	I2S bit clock (BCLK)	Yellow
D8 / GPIO 8	I2S word select (LRC)	Green
D9 / GPIO 9	I2S data (DIN)	Blue
D3 / GPIO 3	Capacitive touch pad	White

PARTS FOR THIS STEP

- **x1** XIAO ESP32S3 Sense, camera attached, pads tinned
- **x5** Jumper wires, cut to **TEMPLE LENGTH + 2CM SLACK** — red, black, yellow, green, blue
- **x1** Jumper wire, white, ~4cm
- **x1** Metal disc — a coin, screw head, or scrap of foil
- **x1** frontboard_base (TPU) and frontboard_lid (PETG)



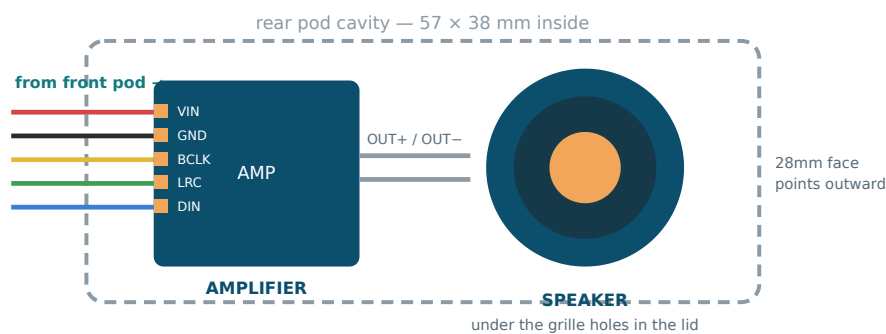
Front pod, viewed from above with the lid off. The camera sits **beside** the board, not on top of it.

- 1 **Cut the five temple wires to length first.** Measure along your actual glasses from the hinge to behind the ear, then add 2cm. Too short is unrecoverable; too long just tucks away.
- 2 Solder red to **3V3**, black to **GND**, yellow to **D7**, green to **D8**, blue to **D9**. Keep the joints flat — anything standing proud eats your lid clearance.
- 3 Solder the white wire to **D3**. This one stays inside the pod.
- 4 Seat the camera in the front of the cavity, lens aligned with the hole in the base. A dab of hot glue holds it.
- 5 Seat the XIAO beside it. **Do not stack the camera on the board** — side by side is what keeps the pod at 10mm and off the wearer's head.
- 6 Route the five temple wires out through the rear-face slot, and the USB cable out the side slot.

Test before you close the lid. Power the board over USB and confirm the camera initialises. Re-opening a hot-glued pod is miserable; catching a dry joint now costs you thirty seconds.

PARTS FOR THIS STEP

- **x1** Audio Converter & Amplifier module (Freenove kit)
- **x1** Speaker, round ~28mm (Freenove kit)
- **x2** Short jumper wires, ~4cm, for the speaker
- **x1** rearaudio_base (TPU) and rearaudio_lid (PETG)
- The five temple wires arriving from the front pod



Rear pod from above, lid off. Amp and speaker sit side by side along the arm.

WIRE IN	FROM (FRONT POD)	TO (AMP)
Red	3V3	VIN
Black	GND	GND
Yellow	D7 / GPIO 7	BCLK (sometimes SCK)
Green	D8 / GPIO 8	LRC (sometimes WS)
Blue	D9 / GPIO 9	DIN

- 1 Feed the five temple wires in through the front-face slot of the rear pod.
- 2 Connect each to the amplifier per the table above. Check the silkscreen labels — some boards print `SCK` for BCLK and `WS` for LRC. They're the same signals.
- 3 Connect the speaker to the amp's two output terminals. **Polarity doesn't matter** with a single speaker.
- 4 Seat the amp and speaker side by side, speaker face pointing outward, under the grille holes in the lid.
- 5 Leave the lid off for now — Step 5 tests audio before you close it.

Why jumper wires and not a ribbon cable? I2S is a slow serial protocol. A 10cm run is electrically fine, and it means this whole build costs nothing beyond what the event handed every team. Say that in the pitch before a judge notices the exposed wires — it reads as a deliberate constraint, not an oversight.

THE TOUCH PAD

There is no touch sensor component. The ESP32's capacitive touch peripheral reads a bare GPIO pin directly — the wire and any conductive surface *are* the whole circuit.

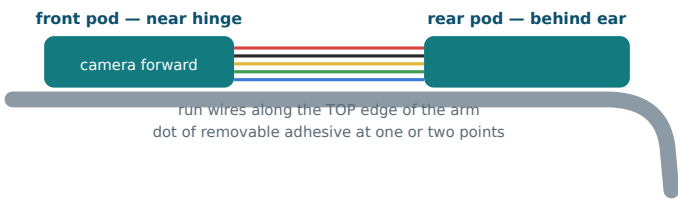
- Glue a coin, screw head, or scrap of foil into the recess on the outside of `frontboard_lid`.
- Solder the white wire from `D3` to it. If it's a coin, scuff the surface first so solder takes.
- Press the lid on. The pad should be reachable with a fingertip when the glasses are worn.

Tune `TOUCH_THRESHOLD` once assembled, not before. The reading changes with the pad's size, the glue, and whether the wearer's head is nearby. Calibrating on the bench and skipping this step is the classic source of a device that false-triggers all through the demo.

POWER

OPTION	HOW	USE IT WHEN
USB-C	Cable out the front pod's side slot to a laptop or power bank	Demo night. Simplest and most reliable — use this.
2×AA holder	~3V to the XIAO's <code>BAT</code> pads	Only if you need untethered. Check voltage first — those pads expect a LiPo.

CLIPPING ONTO THE GLASSES



- Clip the front pod on near the hinge, camera facing forward along the wearer's line of sight.
- Clip the rear pod on behind the ear.
- Run the five wires along the **top** edge of the temple arm, with a dot of removable adhesive so they don't snag.
- Put the glasses on. Check nothing presses into the head and the arm doesn't splay.

Both pods sit beside the arm, not on top of it. That's why they're thin vertically (10mm) and long outward (29–38mm). A temple arm rests against the skull — there's no room above or below it, but plenty straight out. If a pod is touching the wearer's head, something is oriented wrong; check it before gluing.

Do not try to make everything work at once. Every stage you skip here is a bug you'll be hunting at 2am with four variables in play instead of one.

#	STAGE	WHAT PROVES IT WORKS
1	Camera only , on the bench, pods not yet connected to each other	Flash, trigger the touch pad, confirm the server receives a frame at <code>/read_label</code> . No audio, no mic.
2	Audio out , still on the bench, with short wires between the pods	The server's spoken reply plays through the speaker.
3	Full temple-length wires , both pods clipped on	Re-test 1 and 2. A longer run occasionally needs a pull-up tweak — I2S at this length usually doesn't.
4	Microphone in	Record a spoken question, confirm the server transcribes it.
5	Wake word — <i>last</i>	Highest-risk item in the project. The touch pad already gives you a complete working demo without it.

Stage 2 uses short wires deliberately. Prove the audio chain works over 5cm before you commit to a 10cm run threaded along a temple arm. If it breaks after Stage 3, you know it's the cable and not the amp.

TROUBLESHOOTING

SYMPTOM	MOST LIKELY CAUSE
Camera never initialises	PSRAM not set to OPI PSRAM . Check this first, every time.
Camera worked, now doesn't	Ribbon unseated or creased. Re-seat it, latch closed.
No sound at all	Amp not powered — check red to VIN , black to GND . Confirm 3V3 with a meter.
Static, buzzing, or noise only	BCLK and LRC swapped. Yellow → BCLK, green → LRC.
Very quiet but correct audio	Speaker not seated under the lid's grille holes, or DIN loose.
Touch triggers by itself	TOUCH_THRESHOLD needs tuning <i>with the pod assembled and worn</i> .
Touch never triggers	White wire not making contact with the metal disc — scuff and re-solder.
Works on the bench, fails when worn	A temple wire is pulling out. Check strain at both pod exits.
Server reachable, no reply	Venue WiFi. Test on a phone hotspot to isolate it.

Print a spare set of pods before demo night. TPU bases are quick, and a cracked clip an hour before you present is the difference between a demo and an apology.



Master wiring chart

Tear this page out and keep it on the bench

EVERY CONNECTION IN THE BUILD

WIRE	FROM	TO	SIGNAL
Red	XIAO 3V3	Amp VIN	Power
Black	XIAO GND	Amp GND	Ground
Yellow	XIAO D7 (GPIO 7)	Amp BCLK / SCK	I2S bit clock
Green	XIAO D8 (GPIO 8)	Amp LRC / WS	I2S word select
Blue	XIAO D9 (GPIO 9)	Amp DIN	I2S data
White	XIAO D3 (GPIO 3)	Metal disc in front lid	Capacitive touch
Grey ×2	Amp OUT+ / OUT-	Speaker terminals	Analogue out (polarity free)
—	Camera ribbon	XIAO Sense connector	Factory ribbon, unmodified
—	USB-C	Power bank / laptop	Power + flashing

Six wires between the pods' boards, two more inside the rear pod. That's the whole electrical build. Pin assignments match `firmware/src/pins.h` — if you change one, change both.

FINAL PRE-DEMO CHECKLIST

☐ CHECK
Both pods clip on firmly and stay put when the wearer turns their head
Camera points along the wearer's line of sight, not off-axis
Touch pad triggers reliably and doesn't false-fire while worn
Speaker is audible at arm's length in a noisy room
Ten triggers in a row with no failure
Temple wires secured, nothing snagging or pulling at either exit
Tested on the venue network — or a hotspot ready as backup
Spare printed pods in the bag
Demo profile loaded at <code>/setup</code> with synthetic data only
The deliberate-failure beat rehearsed — it says "I can't see that clearly", not a guess

Synthetic data only. The challenge brief is explicit: no real medical or medication information in the prototype. The demo profile is fictional, and saying so plainly on stage is a strength, not a caveat.